Session 5 · Deck 5.B2

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Choosing a Basis: Monomials to Chebyshev

Zhigang Feng

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Where We Are

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Sep 24	Oct 1	Oct 8	Oct 15	Oct 22	Oct 29	Nov 5	Nov 12	Nov 19	Dec 3
Intro	HA · OLG	Num. DP	Numerics	Numerics	HA classic	ML	RL	Text	Agents
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Block B — projection: 5.B1 the residual · **5.B2** choosing a basis · 5.B3 many dimensions and finite elements · 5.B4 coefficients and the worked example.

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Roadmap: Four Parts

5.B1 left the basis unspecified. This deck chooses it, in one dimension: the obvious choice fails on a computer, and one family fixes every problem it has.

Part	Topic	The question it answers
1	Global bases and Gibbs	What is a spectral method, and what goes wrong at a kink?
2	Why not monomials	$1, x, x^2, \ldots$ can approximate anything; why are they unusable?
3	Orthogonality	What does "orthogonal" mean for functions, and what does it buy?
4	Chebyshev	Which orthogonal family, and why is it the default?

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By the end: you will write a policy function as $\sum_i \theta_i T_i(x)$ with Chebyshev polynomials T_i, know the three lines that compute them, and know why no one uses x^i.

Spectral Methods: A Global Basis

A global basis function is nonzero and smooth almost everywhere on the domain. Using one is called a spectral method.

Two kinds of basis — the first of 5.B1's two choices:

- **Global** — each ψ_i is nonzero and smooth almost everywhere on Ψ: **spectral methods**. Polynomials, Fourier series.
- **Local** — each ψ_i lives on a small piece of Ψ: **finite elements** (5.B3). Tent functions, splines.

Idea

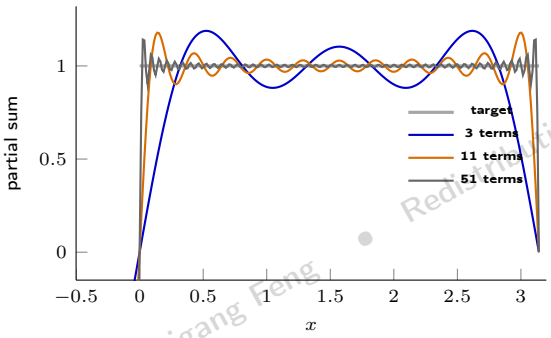
A global basis is like describing a melody by its notes: a few frequencies capture a smooth tune exactly. A sudden jump needs every frequency at once.

Advantage of a global basis: simplicity. One formula holds everywhere. **Disadvantage:** it is hard to capture *local* behaviour. A function smooth here and kinked there must be matched by shapes that are the same everywhere. The symptom is the **Gibbs phenomenon** (next frame).

Main reference: Judd (1992), *Journal of Economic Theory* 58.

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Sines and cosines are the classical global basis. They suit periodic functions, which economics does not have, and at a jump the overshoot never goes away.



The basis $\frac{1}{\sqrt{2\pi}}, \frac{\cos x}{\sqrt{2\pi}}, \frac{\sin x}{\sqrt{2\pi}}, \ldots, \frac{\cos kx}{\sqrt{2\pi}}, \frac{\sin kx}{\sqrt{2\pi}}$ repeats every 2π. A capital stock does not.

Fitted to a square wave, the partial sums overshoot beside the jump, and **more terms do not shrink the overshoot**: 18.84% with 3 terms, 17.90% with 51, converging to 17.9% of the half-amplitude. They only push the ripple closer to the jump.

The cost

At a jump or a kink the error falls only like $1/n$ in the number of terms. For a smooth function the same basis converges far faster. Smoothness is what a global basis is paid for.

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Ledger `basis_conditioning.gibbs`. **Macro has kinks: borrowing limits (2.A1–2.A2), irreversible investment, discrete choice. Those need piecewise bases (5.B3) or learned ones (S7).**

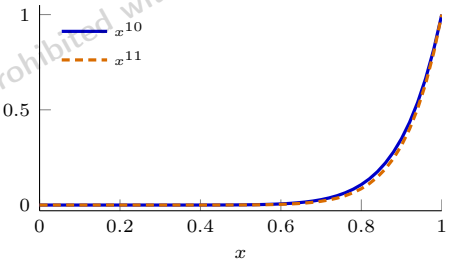
Monomials: The Theory Says Yes

The obvious basis is $1, x, x^2, x^3, \ldots$ A theorem says nothing is missing from it. The practice says two of its members are nearly the same function.

Monomials are simple and intuitive, and the **Stone–Weierstrass theorem** is reassuring: every continuous function on a closed interval $[a, b]$ can be approximated *uniformly* — as closely as you like, at every point at once — by a polynomial.

Problem 1: near-collinearity. Compare the graphs of x^{10} and x^{11} on $[0, 1]$. They are almost the same curve. A fit that must decide how much of *each* to use is being asked a question the shapes can barely answer.

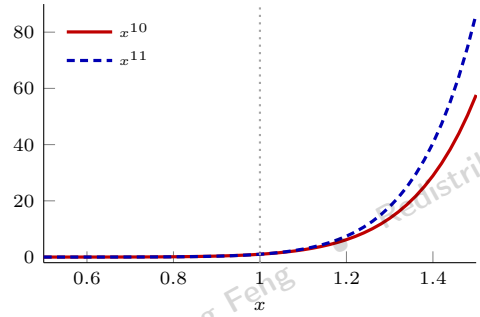
Problem 2: size — next frame.



Approximation theory says the basis is fine. Numerical linear algebra says it is not. The second one wins.

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Monomials vary enormously in size across a domain, so their coefficients do too, and rounding errors accumulate.



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Ledger `restored_figures.monomials_wide`.

The same two monomials on $[0.5, 1.5]$. Below 1 they are indistinguishable; above 1 they run to 58 and 86.

A fit on this domain needs coefficients that matter enormously at one end and hardly at all at the other. They come out at very different sizes, and the rounding errors in the large ones swamp the small ones.

Rescaling the domain to $[-1, 1]$ (Part 4) removes the size problem. It does not remove the near-collinearity — that needs a different basis.

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Where That Matrix Comes From

Least squares with monomials: one first-order condition per coefficient, together a linear system $H\theta = b$.

Fit f on $[0,1]$ with $p(x) = \sum_{j=0}^{n} \theta_j x^j$, choosing coefficients to minimise the squared error:

$$\min_{\theta} \int_0^1 \Big(f(x) - \sum_j \theta_j x^j \Big)^2 dx.$$

Differentiate with respect to **one** coefficient, θ_i, and set to zero — this is **equation** i:

$$\int_0^1 x^i \Big(f(x) - \sum_j \theta_j x^j \Big) dx = 0$$

$$\sum_j \theta_j \underbrace{\int_0^1 x^i x^j dx}_{H_{ij}} = \underbrace{\int_0^1 x^i f(x)\, dx}_{b_i}$$

Do it for $i = 0, \ldots, n$: $H\theta = b$, $n+1$ equations in the $n+1$ coefficients. H_{ij} is **how much θ_j enters the equation for θ_i.**

What H and b hold

H_{ij} is the **overlap of basis function i with basis function j**; b_i is its overlap with the *target*. The system says: give the approximation the same overlap with every basis function that f has.

For monomials, $H_{ij} = \int_0^1 x^{i+j} dx = \frac{1}{i+j+1}$: the **Hilbert matrix**. Every entry lies in $(0,1]$: $i, j \geq 0$, and $0 \leq x^{i+j} \leq 1$ on $[0,1]$. Only $H_{00} = 1$.

b **depends on the function being fitted;** H **does not: what follows is a property of** ***the basis*****, not of the problem.**

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What Near-Collinearity Does to the Equations

The system becomes nearly singular: the solve succeeds, but the numbers it returns are not pinned down.

Neighbouring monomials overlap almost completely. Measured as an angle under the inner product $\int_0^1 fg$, x^{10} and x^{11} are $2.6°$ apart ($\cos = 0.999$): almost the same function. So the Hilbert matrix is **nearly singular**. With nine monomials its condition number is 4.9×10^{11}: the solve loses about 12 of the 16 digits a double carries.

Why. Since x^{10} and x^{11} are almost the same function, raising θ_{10} and lowering θ_{11} leaves the fit almost unchanged. Very different θ fit equally well; rounding decides which one you get.

Idea

It is collinear regressors. Fitted values are fine; the individual coefficients are not identified. A projection method solves for the coefficients and carries them around, so this is fatal.

Why we want $H_{ij} = 0$ for $i \neq j$. Then H is diagonal and equation i holds one unknown: $\theta_i = b_i / H_{ii}$. No system to solve, nothing for rounding to amplify. Basis functions that do not overlap are an **orthogonal basis** — Part 3.

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Ledger `basis_conditioning.hilbert_angles`, `basis_conditioning.hilbert` (9×9).

Orthogonal: Inner Product Zero, and Four Benefits

Two functions are orthogonal when their inner product is zero. With an orthogonal basis the coefficients can be read off one at a time.

For vectors the inner product is the dot product; for functions it is an integral with a weight, $\langle f, g\rangle = \int_{-1}^{1} f\, g\, w\, dx$. A basis is **orthogonal** if $\langle \psi_i, \psi_j\rangle = 0$ for $i \neq j$, **orthonormal** if also $\langle \psi_i, \psi_i\rangle = 1$.

- **Simplicity.** Coordinates are projections onto the basis; no system to solve.
- **Independence.** Orthogonal vectors are linearly independent; each adds a new direction.
- **Numerical stability.** Small changes in the input, small changes in the coefficients.
- **Interpretability.** Each coefficient means something on its own.

Idea

Orthogonal basis functions have *no overlap*: $H_{ij} = 0$ off the diagonal, so the system $H\theta = b$ falls apart into $n+1$ separate one-line equations.

PCA is the same idea on data: axes (the principal components) that are orthogonal, so each is uncorrelated with the others and readable on its own.

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Three lines show why: take the inner product of both sides with one basis vector, and every other term disappears.

In an orthogonal basis $\{v_1, \dots, v_n\}$ write any vector as

$$x = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n.$$

Take the inner product of both sides with v_i:

$$\langle x, v_i \rangle = \sum_{j=1}^{n} c_j \langle v_j, v_i \rangle = c_i \langle v_i, v_i \rangle,$$

since every term with $j \neq i$ vanishes. Solve for c_i:

$$c_i = \frac{\langle x, v_i \rangle}{\|v_i\|^2}, \qquad \text{or simply } c_i = \langle x, v_i \rangle \text{ if orthonormal.}$$

Each coefficient is one inner product. Nothing depends on the other coefficients, so adding a basis function never changes the ones already computed.

For functions

$v_i \to \psi_i$ and $\langle f, g \rangle = \int f g \, w \, dx$:
$c_i = \int f \psi_i w \, dx \,/ \int \psi_i^2 w \, dx$ — Part 2's
$\theta_i = b_i / H_{ii}$ with H diagonal. Chebyshev,
$w = 1/\sqrt{1-x^2}$: $c_i = \frac{2}{\pi} \int f T_i w \, dx$ ($\frac{1}{\pi}$ for $i = 0$).

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Galerkin (5.B4) uses the same idea: a residual orthogonal to every basis function has no component the basis could still represent.

A worked example with numbers: the coordinates of $(3,4)$ in a rotated orthonormal basis are two dot products.

Take $x = (3,4)$ and the orthonormal basis

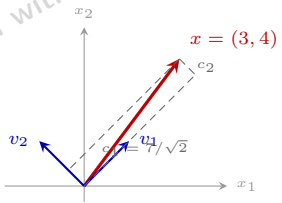
$$v_1 = \left(\tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}\right), \qquad v_2 = \left(\tfrac{-1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}\right).$$

The coordinates are inner products, nothing more:

$$c_1 = \langle x, v_1 \rangle = \tfrac{3}{\sqrt{2}} + \tfrac{4}{\sqrt{2}} = \tfrac{7}{\sqrt{2}},$$
$$c_2 = \langle x, v_2 \rangle = -\tfrac{3}{\sqrt{2}} + \tfrac{4}{\sqrt{2}} = \tfrac{1}{\sqrt{2}},$$

so $x = \tfrac{7}{\sqrt{2}} v_1 + \tfrac{1}{\sqrt{2}} v_2$. **Check:** rebuild x from its coordinates,

$$c_1 v_1 + c_2 v_2 = \left(\tfrac{7}{2} - \tfrac{1}{2},\ \tfrac{7}{2} + \tfrac{1}{2}\right) = (3,4). \checkmark$$



In a non-orthogonal basis those two numbers would need a 2×2 system. With the monomials of Part 2, a system that rounding has already destroyed.

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One Family of Orthogonal Polynomials, Two Famous Members

Orthogonal polynomials are defined by the weight in the inner product. One flexible class, the Jacobi family, contains both members we use.

The **Jacobi** (or hypergeometric) polynomial of degree n, $P_n^{a,b}$, for parameters $a, b > -1$, is defined by the orthogonality condition

$$\int_{-1}^{1} \underbrace{(1-x)^a (1+x)^b}_{\text{weight } w(x)} P_n^{a,b}(x) P_m^{a,b}(x)\, dx = 0, \quad m \neq n.$$

The weight decides which family you get; $a, b > -1$ keeps the integral finite at the endpoints.

- $a = b = -\frac{1}{2}$: **Chebyshev**. The weight blows up at ± 1: extra attention to the ends.
- $a = b = 0$: **Legendre**. Uniform weight; behind the Gauss–Legendre quadrature of 4.B2.

Where a, b enter: only the weight

w defines the inner product; making $1, x, x^2, \ldots$ orthogonal one by one under it (Gram–Schmidt) produces the polynomials. Same monomials, different weight, different family:

$$T_2 = 2x^2 - 1, \qquad P_2 = \tfrac{1}{2}(3x^2 - 1).$$

Check against 1:
$\int_{-1}^{1} \frac{2x^2-1}{\sqrt{1-x^2}} dx = 2 \cdot \frac{\pi}{2} - \pi = 0$;
$\int_{-1}^{1} \frac{1}{2}(3x^2 - 1)\, dx = 0$. From here on only $a = b = -\frac{1}{2}$ is used.

That endpoint weighting is why Chebyshev nodes cluster at the ends of the interval, and why they defeat Runge's phenomenon (4.B1).

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Why Chebyshev Is the Default

One of the most common tools of applied mathematics. Six reasons, and each one answers a problem the earlier parts raised.

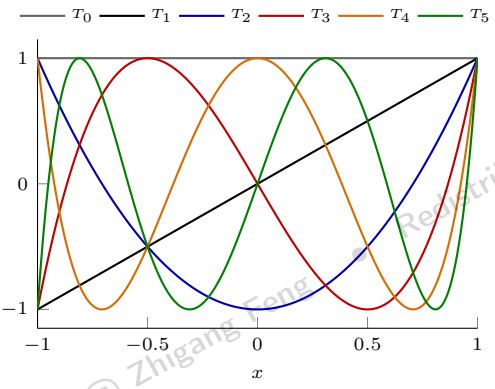
- Numerous **simple closed-form expressions** are available.
- More **robust than the alternatives** for interpolation.
- **Bounded** between $[-1, 1]$ — no coefficient has to cancel a large number, which is what killed the monomials (Part 2).
- **Smooth** functions.
- Moving between the coefficients of an expansion and the function's values at the Chebyshev nodes is a **cosine transform** — fast.
- Several **theorems bound the interpolation error**, so the choice can be defended rather than asserted.

Standard references: Boyd (2001), *Chebyshev and Fourier Spectral Methods*; Fornberg (1998), *A Practical Guide to Pseudospectral Methods*.

A rule of thumb worth carrying: when in doubt, use Chebyshev — unless the problem is periodic, in which case use Fourier.

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Three lines generate the whole family. Every member stays inside $[-1, 1]$, and the zeros crowd toward the ends.



$$T_0(x) = 1, \qquad T_1(x) = x,$$
$$T_{n+1}(x) = 2x\,T_n(x) - T_{n-1}(x),$$

giving $1,\ x,\ 2x^2 - 1,\ 4x^3 - 3x,\ 8x^4 - 8x^2 + 1, \ldots$

The n zeros of T_n are

$$x_k = \cos\frac{2k-1}{2n}\pi, \qquad k = 1, \ldots, n,$$

not equally spaced: they **cluster toward** ± 1, exactly where 4.B1 needed the nodes.

Read the picture. All the same size: every T_n stays in $[-1, 1]$. One more wiggle each: with $x = \cos\vartheta$, $T_n = \cos n\vartheta$ is a cosine of the next frequency.

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Closed Forms, the Change of Variable, and the Theorem

A polynomial written as a cosine; a linear map that puts any interval onto $[-1,1]$; and the theorem that makes the choice safe.

$$T_n(x) = \begin{cases} \cos\left(n \arccos x\right) & -1 \le x \le 1 \\ \cosh\left(n \operatorname{arcosh} x\right) & x \ge 1 \\ (-1)^n \cosh\left(n \operatorname{arcosh}(-x)\right) & x \le -1 \end{cases}$$

The cosine form is why $|T_n| \le 1$ on the interval; outside it the polynomial grows like $\cosh$.

The domain is $[-1,1]$. Our state space is $[a,b]$, so map it in by the linear translation

$$x \longmapsto 2\,\frac{x-a}{b-a} - 1.$$

Chebyshev polynomials are orthogonal with respect to the weight $w(x) = 1/\sqrt{1-x^2}$, infinite at ± 1: the endpoint emphasis again.

Chebyshev interpolation theorem

If an approximating function is exact at the roots of the nth-order Chebyshev polynomial, then as $n \to \infty$ the approximation error becomes arbitrarily small.

Pitfall

Choosing $[a,b]$ is a modelling decision, not a formality: the worked example in 5.B4 takes $[0.5k^*, 1.5k^*]$, and a simulation that leaves it is extrapolating a $\cosh$.

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Takeaways

- **Part 1.** A global basis (spectral method) is simple and converges fast for smooth functions. At a jump or kink it shows the Gibbs phenomenon: the overshoot stays at 17.9% however many terms are added. Kinks need piecewise or adaptive bases.
- **Part 2.** Monomials are complete by Stone–Weierstrass and unusable in practice: x^{10} and x^{11} are nearly the same function, so the coefficient equations (the Hilbert matrix) are nearly singular and rounding decides the answer.
- **Part 3.** Orthogonal means inner product zero. Then each coefficient is one inner product, $c_i = \langle x, v_i \rangle / \|v_i\|^2$: no system to solve, each coefficient meaningful on its own.
- **Part 4.** Chebyshev polynomials T_n: the Jacobi family at $a = b = -\frac{1}{2}$, generated by $T_{n+1} = 2xT_n - T_{n-1}$, bounded, with zeros clustered at the ends; map the state space onto $[-1, 1]$ first. Exact at the roots of T_n means arbitrarily accurate as n grows.

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Bridge: More Than One State Variable

Everything here was in one dimension. A model with capital and a productivity shock has two states; a model with several assets or several agents has many more.

The Chebyshev basis extends to d dimensions by multiplication — and the count of basis functions explodes with d. The next deck shows two ways of keeping it small, and then the opposite choice: a local basis that handles kinks.

Next (5.B3)

Tensor products and the curse of dimensionality; complete polynomials and Smolyak as a *basis* (4.B1 built the grid); finite elements and tent functions; and how to choose between them.

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